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Ornithopter Project Overview

Introduction

My research interest is mechanical design based on kinematics and dynamics. That lead me towards experiences in mechanism study, precision design, compliant mechanism design, etc. Here, I illustrate my design-test-model-optimize procedure with my ornithopter project. An ornithopter is a vehicle that flies forward by flapping its wings. I designed a hundred-gram albatross-scale one due to its payload capacity for potential add-ons (jumping/perching mechanisms for example).

Progress

1. Designed and Tested My 1st Prototype

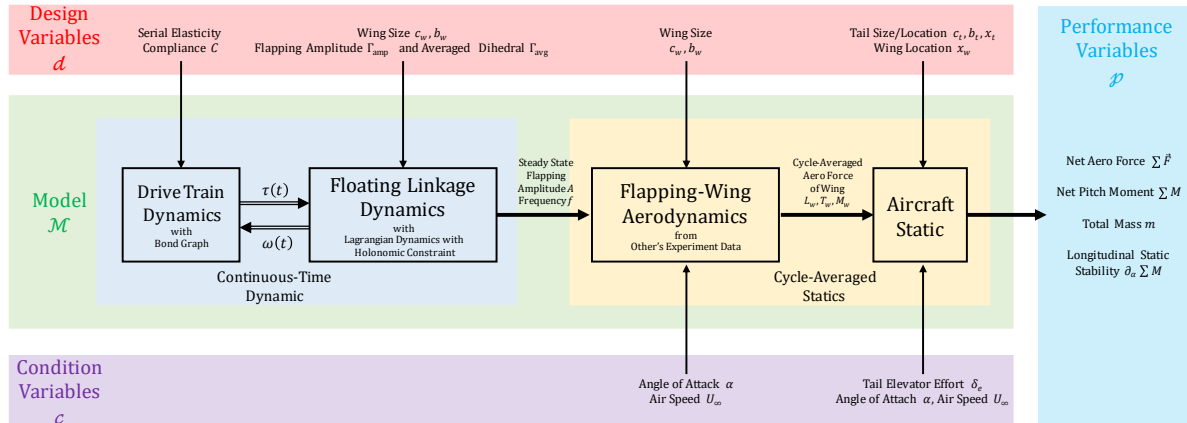
May 2020 to June 2021

I prepared myself for this project with an aerospace degree. I designed my 1st prototype for surviving failures. I innovated a serial-elastic compound gear to protect the drivetrain from crash impact and help motor reduce periodic peak dynamic load. I modularized the design of wings, fuselage and embedded system for easy repairment. I sized the aerodynamics surfaces to ensure stable longitudinal free response and created a simple mechatronics system to further increase stability. A flight test video is [here](#).

2. Created a Multi-Discipline Model and Achieved Optimization

Dec. 2021 to May 2022

I directed my coursework to achieve this analytical approach, so as to get some deterministic light to guide me through my design process. My model $p = \mathcal{M}(d, c)$ innovatively coupled the electro-mechanical and aero-aircraft aspects to predict flight performance in a computationally affordable manner. With that, I created an optimization code and found the design d that minimized takeoff-speed.



Closing Remark

I may one day continue the ornithopters if time allows, but I am more than willing to devote myself to any other designs projects as well. Ornithopters are dear to me, but dearer still is the general mechanical design based on kinematics and dynamics.

Acknowledgement

Please visit [this page](#) for the list of acknowledgements. I launched the ornithopter project purely out of passion, and I couldn't have made it without people's help.